Conservation of Energy

$$P_f = P_i$$

(final Power) = (initial power)

$$P_r + P_t = P_i$$

P_t: Power of transmitted wave

P_i: power of incident wave

P_r: Power of reflected wave

$$\left[\frac{P_r}{P_i} + \frac{P_t}{P_i}\right] = 1$$

$$\underbrace{\frac{P_r}{P_i} + \frac{P_t}{P_i}} = r^2 + \frac{Z_2}{Z_1} t^2$$

$$= \left[\frac{Z_2 - Z_1}{Z_1 + Z_2}\right]^2 + \frac{Z_2}{Z_1}\left[\frac{2Z_1}{Z_1 + Z_2}\right]^2$$

$$= \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2} + \frac{4 Z_1 Z_2}{(Z_1 + Z_2)^2}$$

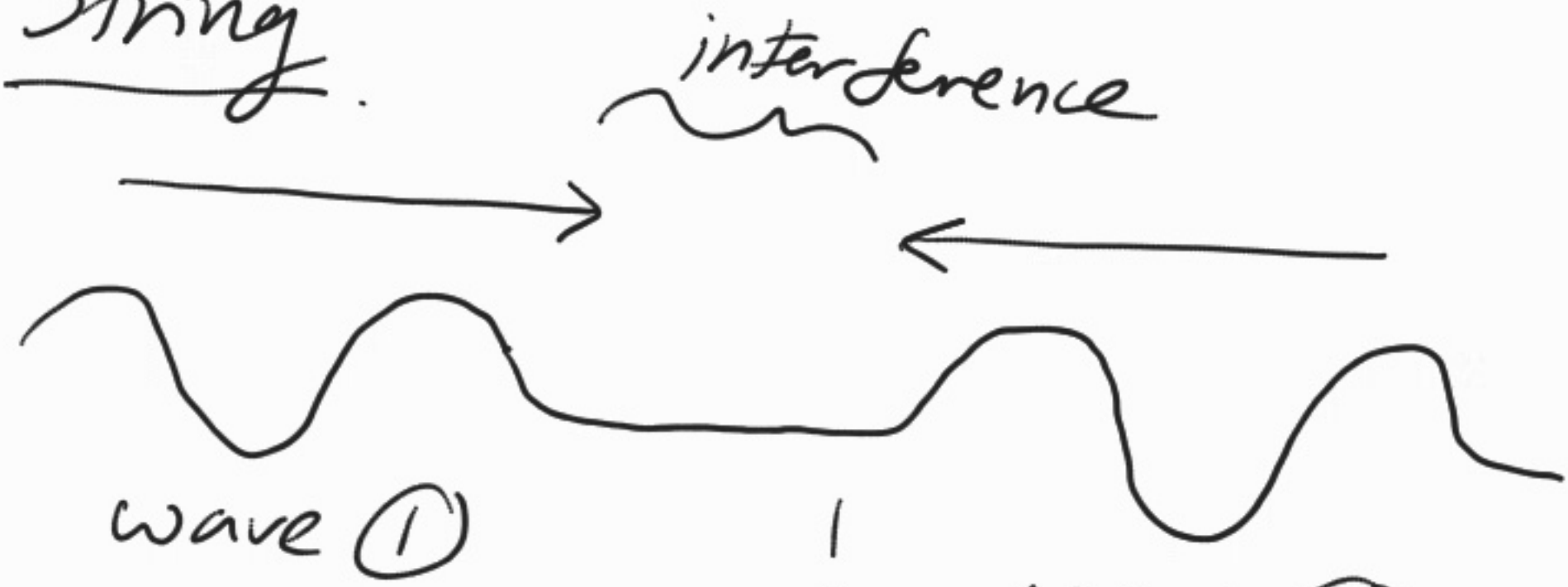
$$= \frac{Z_1^2 + Z_2^2 + 2 Z_1 Z_2}{(Z_1 + Z_2)^2} = 1$$

$$T + R = 1$$

Note:

$r + t \neq 1$

Standing waves on a Stretched String.



$$y_1 = y_m \sin(kx - \omega t)$$

$$y_2 = y_m \sin(kx + \omega t)$$

Resultant wave: $y_R = y_1 + y_2$

$$\sin\alpha + \sin\beta = 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha+\beta}{2}\right)$$

∴ Standing wave :

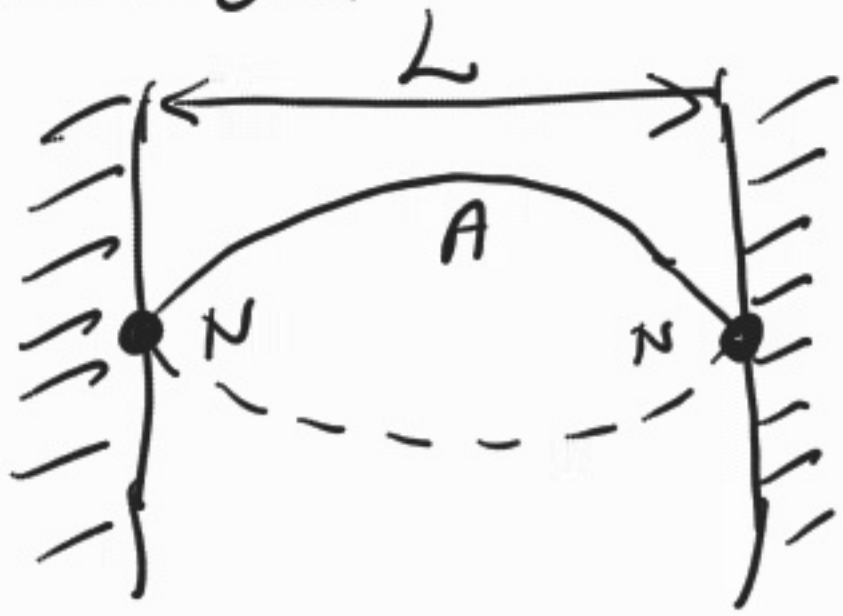
$$y_R = 2y \sin(kx) \cos(\omega t)$$

$2y \sin(kx)$ → Standing wave amplitude → at a given time the shape of the string is a sine curve

$\cos(\omega t)$ → the shape of the wave stays in the same position : oscillating up ↑ & down ↓.

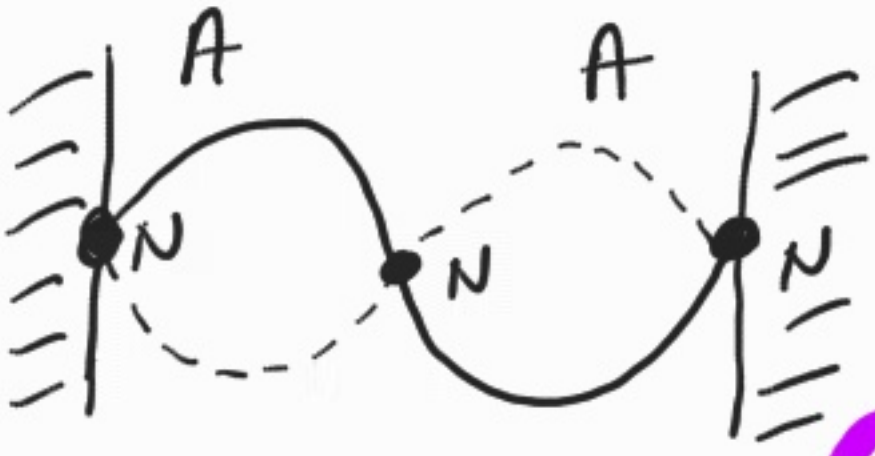
Note :
This is not a travelling wave since it is not in the form of $\sin(kx - \omega t)$

Standing Waves on A Stretched String.



1st Harmonic

$L = \lambda/2$

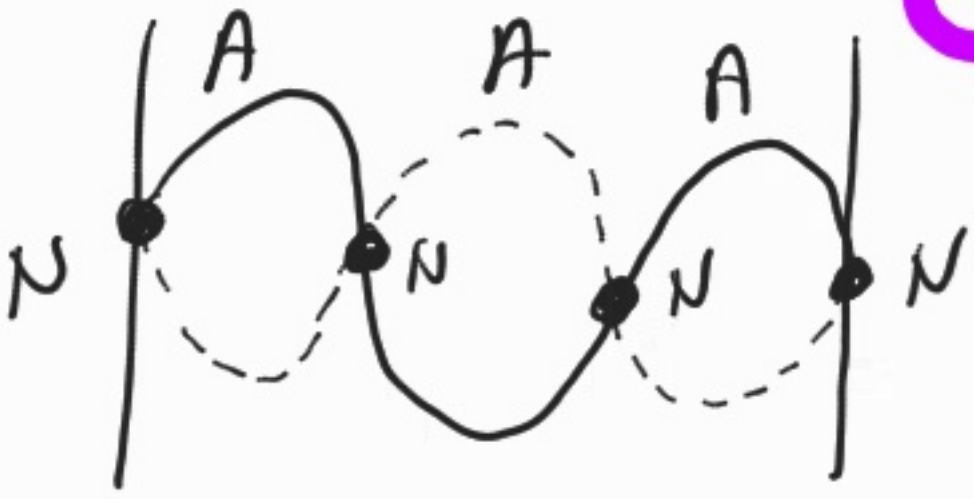


2nd Harmonic

$L = \lambda$

N: node, zero-amplitude

A: antinode, max amplitude



3rd Harmonic

$L = 3\lambda/2$

Standing Waves

- Possible standing waves have the wavelengths of:

$$\lambda = \frac{2L}{n}, \quad n = 1, 2, 3, \ldots$$

- Resonant Frequencies:

$$f = \frac{v}{\lambda} = n\frac{v}{2L}$$

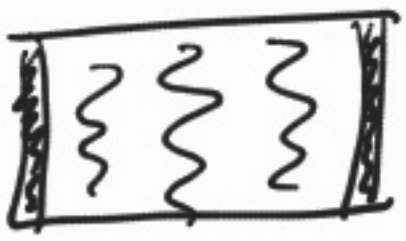
- Fundamental mode:

$n = 1$ where $f = \frac{v}{2L}$

Standing Sound Waves

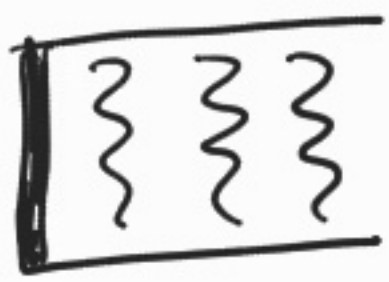
Case 1

Both ends Closed



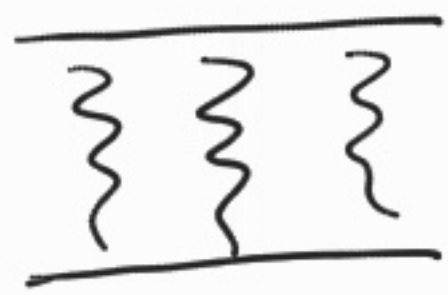
Case 2

One end open



Case 3

Both ends open



Types of Harmonics

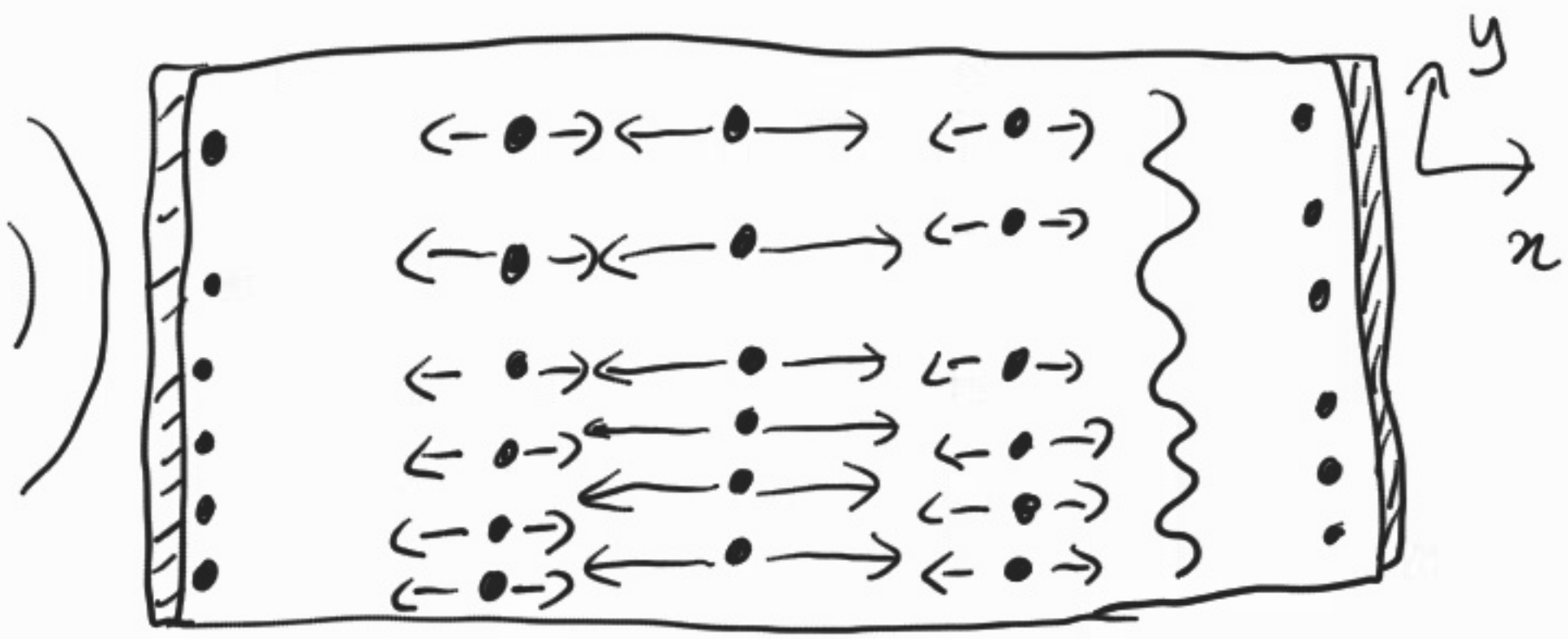
i) 1st Harmonic

ii) 2nd Harmonic

iii) 3rd Harmonic

Standing Sound Waves

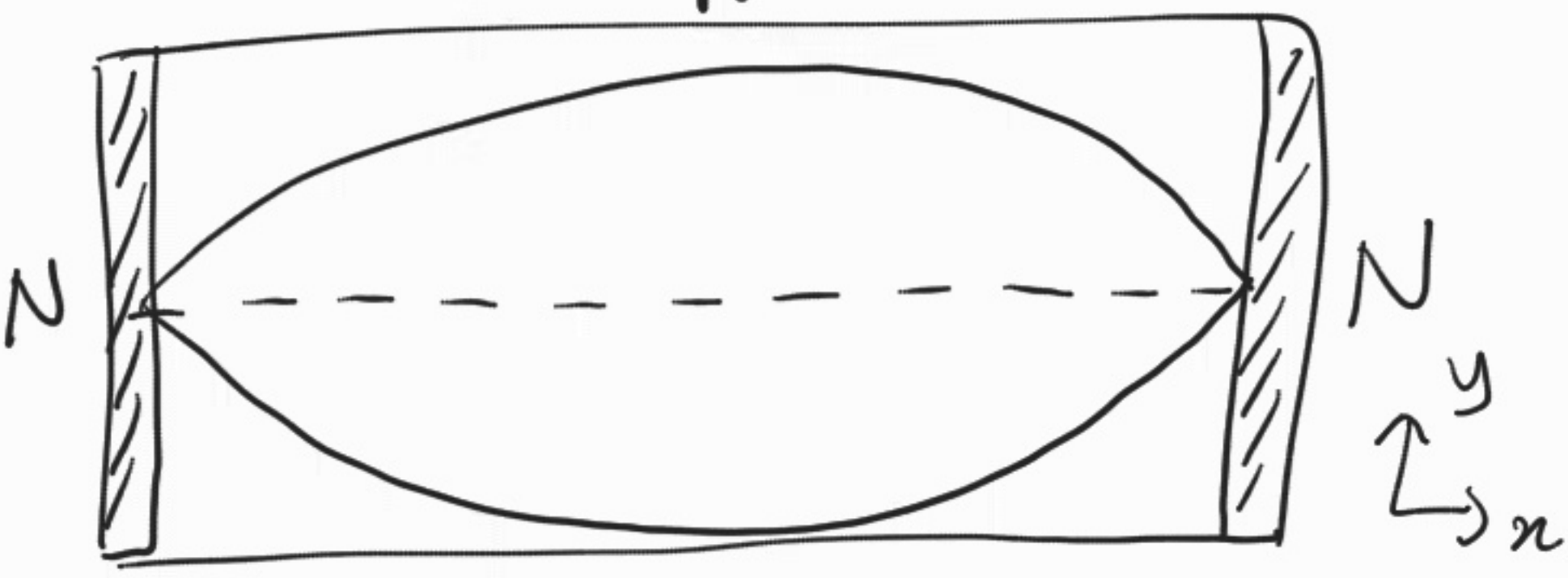
- Pipe with closed ends (1st harmonic)



longitudinal oscillation of air molecules

Standing Sound Waves

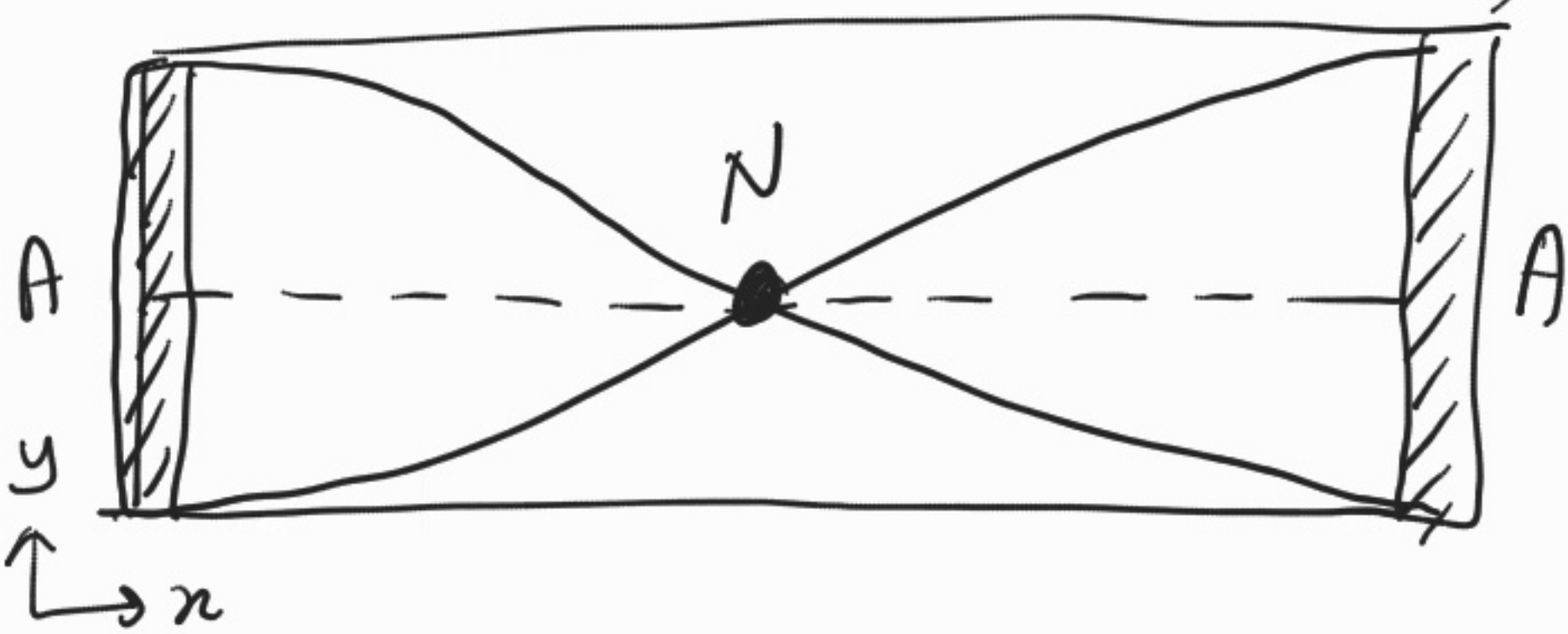
A (1st Harmonic)



- The displacement at both ends of the pipe is zero and maximum in the middle.

Standing Sound Waves

(1st Harmonic)



- The pressure at both ends of the pipe is at max & is zero in the middle.

Note:
Pressure node is always a displacement antinode and vice versa